

High-Fidelity Five-View Jewellery Reconstruction

Feasibility, open-source model stack, implementation status and research roadmap

Report date: 28 September 2026

Project checkpoint reviewed: Ring01 Phase 3.3.2

Input scenario: one to five reference images, with special attention to front, back, side, top and angled views

Target output: visually faithful, component-aware and editable 3D/CAD with explicit uncertainty

2D JEWELLERY TO EDITABLE 3D/CAD



Executive decision

The project can build a strong five-view reconstruction system with open-source computer vision, depth, correspondence, inverse-rendering and CAD tools. The correct target is **near-exact visible-view agreement with editable structure**, followed by metric and manufacturing validation when physical evidence is available.

A mathematically perfect replica cannot be recovered or verified from five ordinary photographs alone. The images do not uniquely determine hidden surfaces, true thickness, physical scale, material reflectance or internal construction. Polished metal adds view-dependent reflections; gemstones add refraction and internal reflection; shadows and highlights can look like geometric edges. Several different 3D objects can therefore explain the same five images.

This does not make the project impractical. It changes the engineering strategy:

1. preserve the original image evidence;
2. classify rather than blindly trace edges;
3. estimate cameras, depth and normals with multiple independent methods;
4. fuse only evidence that agrees across views;
5. represent regular jewellery as exact parametric CAD and irregular detail as constrained free-form surfaces;
6. re-render every candidate into all source cameras;
7. expose uncertainty instead of inventing unsupported detail;
8. require physical scale or scan evidence before claiming manufacturing accuracy.

The existing repository already implements the evidence, segmentation, visual-hull, exact-solid and validation foundations needed for this direction. The next highest-value work is not another general image-to-3D generator. It is a calibrated camera/depth benchmark, human-reviewed component truth, edge taxonomy, uncertainty-aware fusion and hybrid CAD/free-form reconstruction.

1. What “perfect texture and edge replication” must mean

An image contains different kinds of boundaries. Treating all of them as geometry creates spikes, false grooves and melted surfaces.

Image signal	Meaning	Correct treatment
Outer silhouette	Object-to-background boundary	Strong geometric constraint in that camera view
Occlusion boundary	One component crosses another	Geometry and depth-order constraint
Crease or ridge	Real surface direction change	Candidate curve, normal or relief constraint
Engraved/embossed boundary	Fine surface geometry	Multi-view relief or displacement constraint
Material boundary	Gold, enamel, stone or paint change	Texture/material mask unless depth evidence agrees
Specular highlight	Reflection of a light source	Lighting/BRDF evidence; never geometry by itself
Cast or contact shadow	Lighting and spatial relation	Weak depth cue; exclude from the object silhouette
Gemstone facet line	Geometry mixed with reflection/refraction	Stone-cut model plus multi-view facet evidence

The target should therefore be split into three deliverables:

- **Base geometry:** watertight shank, body, settings, galleries, links and cavities.
- **Microgeometry:** engraving, relief, faces, motifs, milgrain, ridges and controlled displacement.
- **Appearance:** base colour, metalness, roughness, normal/displacement maps and view-dependent material behaviour.

A colour texture alone cannot reproduce polished gold. A mesh alone cannot reproduce enamel and patina. The final system needs physically based materials and geometry, with the source images used to distinguish them.

2. Is five-view reconstruction achievable?

2.1 What five good views can constrain

Five views can be useful when they show the **same physical object**, retain full resolution, overlap substantially and use known or recoverable cameras. They can constrain:

- principal silhouettes;
- visible component count and placement;
- broad depth ordering;
- front/back/side proportions;
- repeated symmetry where it is genuinely present;
- stone and prong locations;
- visible negative spaces;
- medium-scale relief seen in more than one view;
- a plausible component-aware CAD model.

2.2 What five ordinary views cannot prove

- hidden surfaces that appear in no image;
- absolute millimetres without at least one known measurement;
- wall thickness and internal cavities that are not visible;
- the true shape under a strong reflection or shadow;
- exact gemstone pavilion, seat and clearance when obscured by metal;
- the physically correct BRDF, illumination and colour calibration;
- one-to-one correspondence when independently generated/reference views contain inconsistent details;
- manufacturing tolerances.

COLMAP's own capture guidance expects overlapping images, similar illumination, sufficient texture and recommends that each object point be seen in at least three images; it also warns against specularities. That is directly relevant to jewellery and explains why five orthogonal catalogue renders are harder than a dense turntable capture: [COLMAP capture guidance](#).

2.3 Practical evidence rating

Target	Five ordinary images	Five calibrated studio images	Dense calibrated capture or scan
Visible silhouette	Strong	Very strong	Very strong
Component layout	Moderate to strong	Strong	Very strong
Camera pose	Weak to moderate	Strong	Very strong

Target	Five ordinary images	Five calibrated studio images	Dense calibrated capture or scan
Broad visible depth	Moderate	Strong	Very strong
Fine relief	Weak	Moderate	Strong with macro views
Hidden/backside geometry	Weak or absent	Weak if still hidden	Strong if covered
Absolute dimensions	Not recoverable	Recoverable with known scale	Recoverable
Metal/enamel appearance	View-specific approximation	Good with controlled lighting	Strong with material capture
Gemstone interior	Unreliable	Still difficult	Requires specialised capture/model
Manufacturing accuracy	Not defensible	Partial after CAD checks	Defensible after scan/CAD inspection

Decision: five views are sufficient for the next research stage, but they are not a universal evidence ceiling. The pipeline should accept five views and report coverage gaps; the capture protocol should request additional macro, underside and oblique views when confidence is low.

3. Recommended open-source depth and geometry stack

No single model should be treated as ground truth. The strongest design is an ensemble in which classical geometry, learned multi-view geometry, monocular depth and surface normals cross-check one another.

3.1 Priority models for the first benchmark

Model or algorithm	Role in this project	Why it is useful	Limitation or caution	Priority
COLMAP	Camera intrinsics/ extrinsics, sparse SfM, classical MVS baseline	Established, inspectable and reproducible; supports SIFT, learned features and LightGlue paths	Shiny, texture-poor jewellery and only five low-overlap views can fail	P0
VGGT	Joint camera, depth, point-map and track proposal	Designed for one, few or many images and supplies several geometric outputs together	Research licence; benchmark on jewellery before depending on it	P0 remote
Depth Anything 3	Spatially consistent multi-view depth and pose proposal	Accepts arbitrary views with or without known poses and exports depth/camera/3D forms	Newer stack; author benchmarks are not project validation; verify every checkpoint licence	P0 remote
MASt3R / DUST3R	Dense cross-view matches and point maps when feature matching is weak	Useful fallback for wide viewpoint changes and weak local texture	DUST3R code is non-commercial CC BY-NC-SA; MASt3R checkpoint terms also require review	P0 research
Depth Pro	Sharp monocular metric-depth and focal-length hypothesis	Explicitly targets sharp high-frequency depth boundaries and predicts focal length	Single-view depth remains a prior, not multi-view truth	P1

Model or algorithm	Role in this project	Why it is useful	Limitation or caution	Priority
Depth Anything V2	Lightweight relative-depth prior	Small model is practical locally and useful for ranking near/far structure	Relative depth must be aligned across views and to known scale	P1 local
DSINE	Per-pixel surface-normal proposal	Crisp, piecewise-smooth normal estimates can protect relief and curved metal	Normals inferred from one image may follow lighting errors	P1
OpenMVS	Dense depth fusion, meshing, refinement and texture mapping	Provides a classical end-to-end dense reconstruction baseline after camera recovery	Needs reliable calibrated views and careful licence review for distribution	P1

Depth Anything 3 states that it predicts spatially consistent geometry from arbitrary visual inputs with or without known camera poses. Its published claims make it worth benchmarking, but those claims must be verified on the project's jewellery and sculptural assets rather than accepted as a guarantee: [DA3 official repository](#). VGGT similarly predicts cameras, depth, point maps and tracks from few or many views and is a strong independent branch for the ensemble: [VGGT CVPR 2025 paper](#).

3.2 Reflective-object and inverse-rendering research

Jewellery is not a normal photogrammetry subject. The system needs a branch that separates geometry from reflections and materials.

Project	Proposed use	Status
NeRO	Joint neural geometry and BRDF reconstruction for reflective objects	High-value research benchmark; expects a richer multi-view capture than five sparse catalogue views
nvdiffrrec	Joint topology, material and lighting optimisation	Useful for learning whether a mismatch is geometry or appearance; remote GPU recommended
nvdiffrast	Low-level differentiable rasterisation	Strong optimisation backend for camera, mesh and texture losses
PyTorch3D	Differentiable render-and-compare, mesh losses and batched cameras	Good integration target for the existing validation loop
Nerfstudio	Modular NeRF/3DGS experiments and view synthesis	Research evidence, not automatically editable CAD
Neuralangelo	Detailed neural surface baseline from dense image/video capture	Better suited to many frames and remote compute than five sparse views

NeRO is especially relevant because it explicitly targets reflective-object geometry and BRDF reconstruction from multi-view images: [NeRO official implementation](#). nvdiffrrec jointly optimises topology, materials and lighting from multi-view observations: [nvdiffrrec official implementation](#).

These methods should not replace the CAD branch. They should supply surface and material evidence that is converted into editable parts or bounded relief patches.

3.3 Segmentation, instances and edge evidence

Model or algorithm	Use
OpenCV Canny, contours, morphology and line/curve fitting	Deterministic silhouette, holes, symmetry, ridge candidates and pixel measurements

Model or algorithm	Use
SAM 2	Promptable jewellery/component masks and cross-frame propagation when a turntable video is available
HQ-SAM / HQ-SAM 2	Boundary-refinement benchmark for prongs, stones, chains and engraved regions
Grounded SAM	Text-guided proposals for stones, prongs, settings, links and motifs
TEED	Lightweight learned edge proposal alongside deterministic Canny
Jewellery-specific instance model	Final per-stone/per-prong IDs after human-reviewed labels exist

SAM-family masks are proposals, not measurements. The output must preserve uncertainty bands and allow reviewer correction. For a ten-stone item, the evidence graph must contain ten stable stone instances, not one merged “stone” mask.

3.4 CAD and geometry processing

- [CadQuery](#) with OpenCascade remains the exact-solid and STEP path for shanks, bezels, prongs, galleries, seats, cavities and assemblies.
- Blender can handle remeshing, sculptural patches and Boolean experiments; its Boolean modifier supports union, intersection and difference operations: [Blender Boolean manual](#).
- Trimesh remains useful for mesh inspection, connected components, watertightness and export checks.
- A robust mesh Boolean backend such as Manifold can be evaluated for cavity and seat generation before conversion/validation.
- UV unwrapping and view-weighted texture baking should create base-colour, roughness, metalness, normal and displacement maps separately.

CadQuery is appropriate because it is a scriptable parametric CAD system based on OCCT and supports STEP, STL, 3MF and other outputs: [CadQuery official repository](#).

4. Proposed depth-estimation architecture

“Almost perfect depth” should be approached as an evidence-fusion problem, not a model-selection problem.

Stage A — camera hypotheses

Run three independent branches:

1. COLMAP with SIFT, then ALIKED/LightGlue if available;
2. VGGT camera and track prediction;
3. DA3 or MAST3R camera/point-map prediction.

Align all successful camera sets into one coordinate frame. Reject solutions with incorrect view ordering, mirrored geometry, implausible focal length or poor source-view reprojection.

Stage B — depth and normal hypotheses

For every view, produce:

- multi-view depth from DA3/VGGT or classical MVS;
- monocular depth from Depth Pro and Depth Anything V2;
- normals from DSINE;
- silhouette and component masks;
- per-pixel confidence and an invalid/reflection mask.

Monocular estimates may fill weak areas, but they must not override cross-view geometry where multiple cameras agree.

Stage C — uncertainty-aware fusion

Convert each valid pixel into a camera ray and fuse only mutually consistent measurements. Each sample receives a weight based on:

- number of supporting views;
- reprojection error;
- edge type;
- depth-model agreement;
- normal agreement;
- segmentation confidence;
- reflection/shadow probability;
- distance from an occlusion boundary.

The result should be a signed-distance/occupancy field plus an uncertainty volume, not only a point cloud. Hidden regions remain explicitly uncertain.

Stage D — hybrid reconstruction

- Fit analytic or parametric components where jewellery structure is known.
- Fit free-form subdivision/NURBS or dense mesh patches for faces, deities, organic motifs and ornament relief.
- Store detail as real displacement when it affects silhouette or manufacture; otherwise retain it as normal/appearance detail.
- Generate gemstone seats and cavities directly from each named stone's geometry and pose.

Stage E — differentiable render-and-compare

Render geometry, depth, normals, component IDs and physically based appearance from the source cameras. Optimise in a coarse-to-fine order:

1. camera and scale;
2. silhouette and component placement;
3. depth and normals;
4. visible creases and relief;
5. materials and texture.

Geometry must be frozen or strongly regularised while optimising appearance so that a highlight cannot carve a groove into the CAD model.

5. Texture and fine-detail replication plan

5.1 Texture is a multi-map result

The texture deliverable should contain:

- base colour/albedo;
- metalness or material class;
- roughness;
- normal map;
- displacement/height where supported;
- optional ambient-occlusion preview map;
- source-view visibility and confidence atlas.

Images should be exposure- and white-balance-normalised before baking. For each texel, choose source pixels using camera angle, focus, occlusion, highlight probability and cross-view consistency. Blend only compatible observations.

5.2 Metal and gemstone rules

- Polished-metal highlights are view-dependent and should be fitted as illumination/BRDF, not copied permanently into albedo.
- Enamel, patina and painted regions belong in material masks and base colour.
- Repeated stamped or engraved motifs should use geometry when they affect silhouette/shadows or when manufacturing requires them.
- Gemstones should use a parametric cut with material shaders; photographed sparkle should not be baked as fixed white spots.
- If a facet is visible in only one view, record it as low confidence until the known cut topology or another view supports it.

5.3 Controlled capture upgrade

Five reference images remain supported, but the preferred research capture should add:

- fixed focal length and locked focus/exposure/white balance;
- a scale marker or one measured object dimension;
- same object and unchanged articulation in every view;
- neutral matte background;
- 12–36 overlapping turntable views for geometry;
- macro detail views for engraving/faces;
- cross-polarised diffuse-light pass to reduce specular glare;
- separate reflective-light pass for material estimation;
- underside and cavity coverage;

- optional turntable angle metadata.

This capture change is likely to improve accuracy more reliably than replacing one large foundation model with another.

6. What has already been implemented

The following results are repository evidence, not projections.

6.1 Input and evidence handling

- Upload audit accepts single-view and five-view inputs.
- Source files, hashes, view labels, quality warnings and transforms are preserved.
- Phase 1 extracts foreground, contours, holes, edges, symmetry and pixel-space measurements.
- Phase 2 combines SAM 2.1 Tiny, OpenCV and GrabCut for silhouette/component proposals.
- Phase 2.2 applies topology and cross-view consistency rules.
- Phase 2.3 stores an evidence graph, instance hypotheses and review decisions.

6.2 Dataset processing

The current dataset contains 24 objects and five views per object, for 120 processed images. The small U-Net experiment reports:

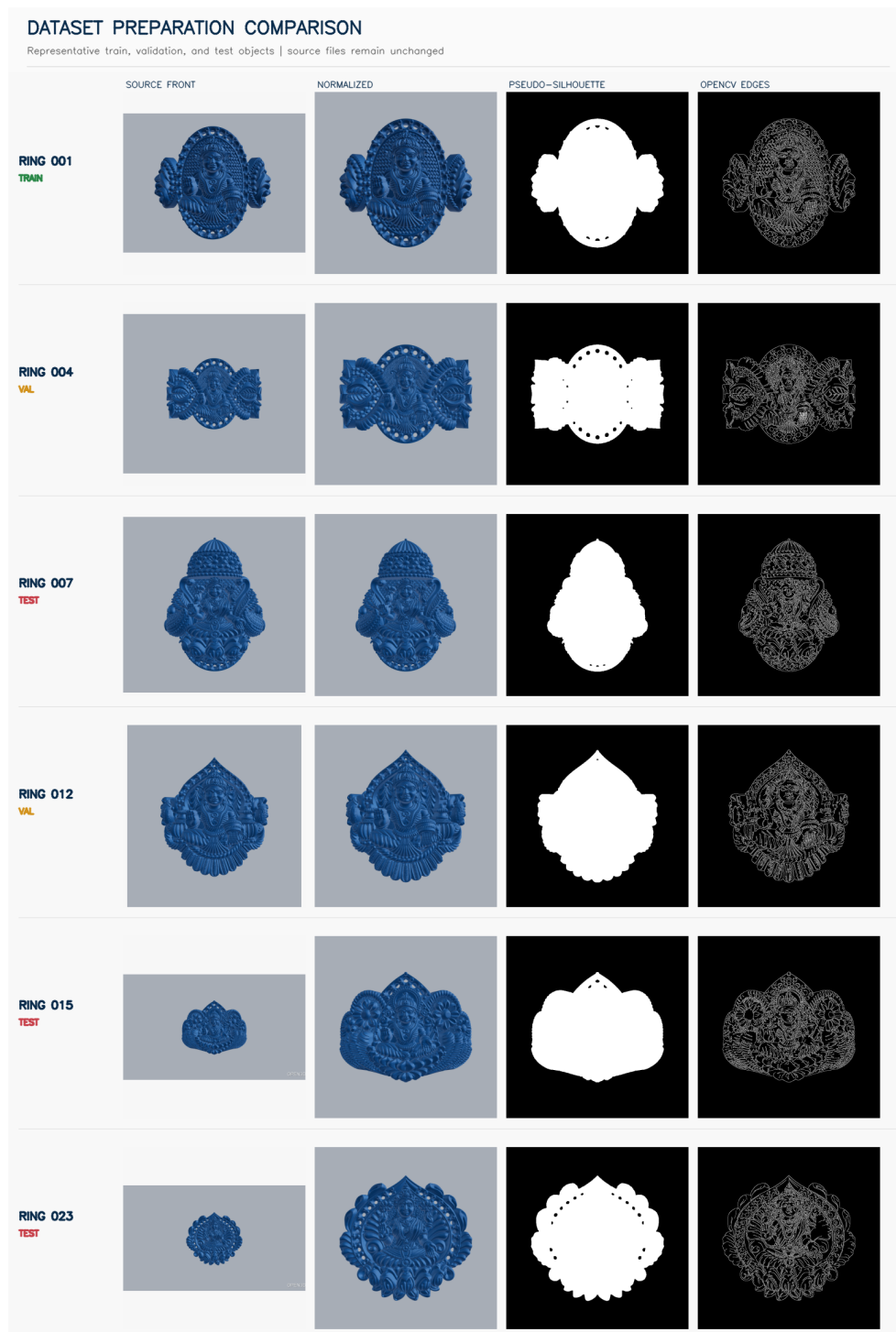
Dataset result	Recorded value
Validation pseudo-mask IoU	0.978183
Held-out pseudo-mask IoU	0.977586
Held-out Dice	0.988646
Held-out boundary F1 at 2 px	0.999753

These numbers measure reproduction of machine-generated pseudo-masks. They do **not** prove human-quality component segmentation, depth accuracy or 3D accuracy.

The dataset visual-hull experiment created 24 watertight coarse meshes at 128-voxel resolution:

Split	Reprojection IoU
Train	0.948832
Validation	0.946122
Test	0.961753

These are non-metric visual-hull reprojection scores. They validate the batch workflow, not hidden shape or manufacturing geometry.



6.3 Ring01 exact reconstruction checkpoint

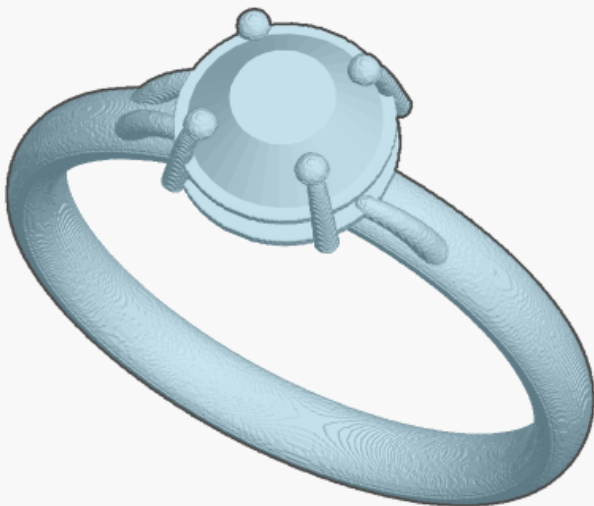
Ring01 Phase 3.3.1 produced one valid metal B-rep, one separate gemstone, a two-rail gallery, shank/shoulders and four named curved cubic claws. Phase 3.3.2 corrected the inspection preview so all four prongs are visible without changing geometry.

Metric	Recorded value
Mean silhouette IoU	0.819726
Mean detail-region IoU	0.787388
Mean external-boundary F1 at 2 px	0.795969

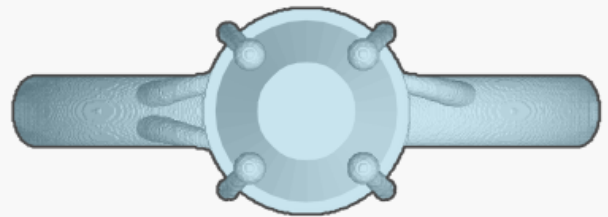
Metric	Recorded value
Front IoU	0.854535
Side IoU	0.813976
Top IoU	0.871267
Angled IoU	0.770454
Back IoU	0.788398

The exact checkpoint is structurally useful, but it has not reached the 0.90 research target, the complete Phase 3 exit gate, or manufacturing validation. Human-reviewed masks and metric scale remain missing.

ISOMETRIC (FITTED) – 4 CLAWS



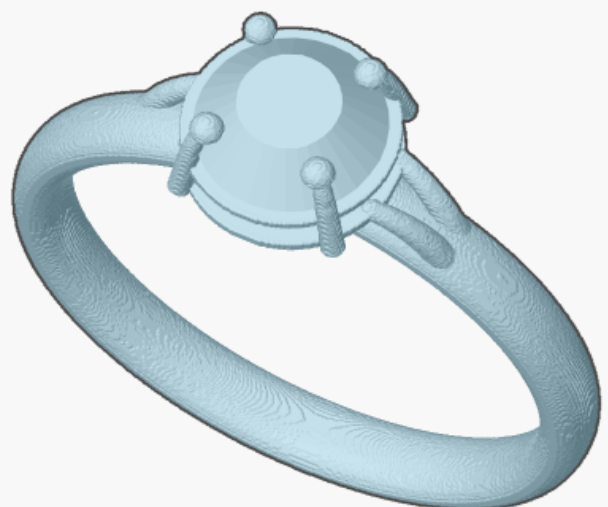
GEM FACE – 4 CLAWS



HOOP PROFILE



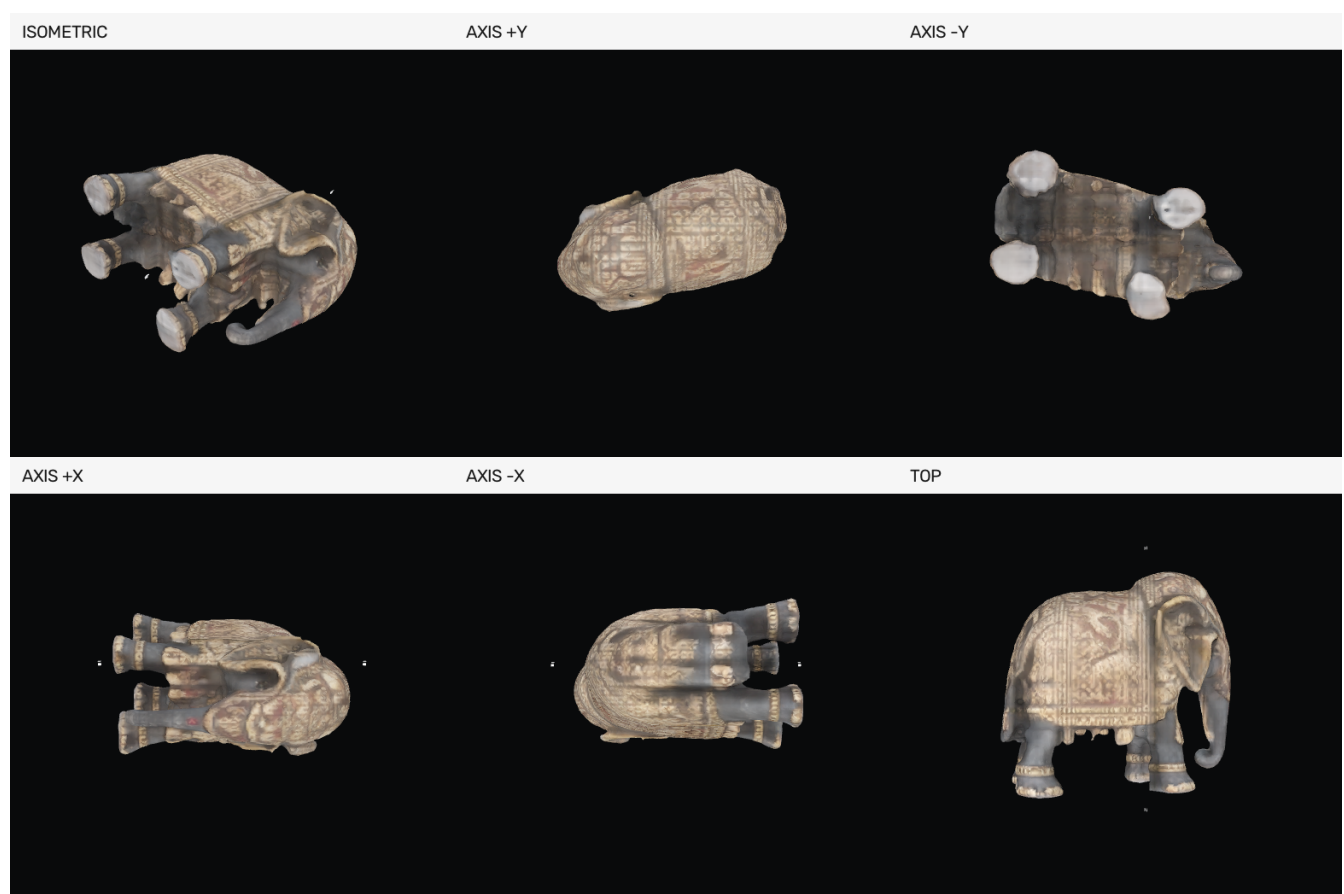
OPPOSITE – 4 CLAWS



6.4 External image-to-3D benchmark findings

The benchmark validates exported geometry using independent renders before metrics.

- **TripoSR**: produced a recognisable elephant, but geometry and texture were too coarse for the ornamental target.
- **Stable Fast 3D**: hosted runs failed before a usable mesh was produced.
- **InstantMesh**: produced a coherent recognisable elephant and passed the human preview gate as research evidence; the mesh was non-watertight and split into 18 connected bodies, so it was not a CAD or manufacturing result.
- **Hunyuan3D 2.1**: exported geometry was catastrophically torn/fused and was rejected at the preview gate.
- **TRELLIS**: the official hosted environment failed at configuration level.
- **TRELLIS.2**: reached a running service, but the free hosted quota blocked inference; no result and no paid transaction were accepted.



The lesson is that generative models are useful candidate generators, not geometric authorities. A visually attractive website preview is not sufficient; the actual OBJ/GLB must be independently rendered, inspected and validated.

7. Gap analysis

Required capability	Current position	Main gap
Whole-object silhouette	Working	Human-reviewed truth at scale

Required capability	Current position	Main gap
Individual stone/prong segmentation	Proposed and partly represented	Jewellery instance dataset and stable cross-view IDs
Camera recovery	Basic assumptions/experimental paths	Calibrated benchmark across COLMAP, VGGT, DA3 and MAST3R
Depth estimation	Visual hull/coarse semantics	Multi-model depth/normal ensemble with uncertainty
Exact parametric body	Ring01 prototype	Universal component grammar across jewellery classes
Stone cavity/seal update	Design requirement	Dependency graph and robust Boolean/clearance engine
Faces/idols/relief	Evidence extraction planned	Free-form surface branch and macro-detail capture
Texture/material	Basic external-model textures	View-weighted PBR baking and reflective-object inverse rendering
Metric validation	Not available from current dataset	Known dimensions, camera calibration and scan/CAD ground truth
Manufacturing claim	Not validated	Rules, tolerances and independent CAD/scan inspection

8. Implementation roadmap

Milestone A — freeze ground truth and capture protocol

Deliverables

- Human-reviewed silhouettes and component masks for Ring01, one multi-stone item and one sculptural/face item.
- Stable IDs for every stone, prong, cavity and visible motif.
- Camera calibration board procedure and one known metric dimension.
- Dense turntable capture for at least one benchmark object, while retaining the five-view subset for comparison.

Exit gate

- Two reviewers agree on instance count and masks, or disagreements are recorded.
- All reference views are the same physical object.
- Ground-truth source is labelled `human_reviewed`, `scan`, `CAD` or `measured`; pseudo-truth is never mixed with it.

Milestone B — edge taxonomy and evidence confidence

Deliverables

- Canny + TEED proposal comparison.
- Boundary classes: silhouette, occlusion, crease/relief, material, highlight and shadow.
- Reflection/shadow confidence map.

- Cross-view edge tracks and uncertainty bands.

Exit gate

- Human-reviewed external-boundary F1 and per-class precision/recall reported.
- Highlight-only boundaries do not modify geometry in regression tests.

Milestone C — camera, depth and normal benchmark

Deliverables

- Isolated adapters for COLMAP, VGGT, DA3, MAST3R/DUST3R, Depth Pro, Depth Anything V2 and DSINE.
- Common NPZ/JSON contract for camera, depth, normals, confidence, runtime and memory.
- Five-view and dense-view runs on the same benchmark objects.
- Licence manifest for code and weights.

Exit gate

- Camera reprojection error, depth error against scan/CAD, normal angular error and failure rate are reported separately.
- No model is selected using its own preview or confidence alone.
- The chosen ensemble beats every single branch on the held-out objects or is rejected.

Milestone D — uncertainty-aware fusion and hybrid geometry

Deliverables

- Calibrated TSDF/SDF or occupancy fusion.
- Uncertainty volume and visible/hidden surface labelling.
- Parametric fitting for regular parts and free-form patches for sculptural detail.
- Source-view correspondence retained for every fitted region.

Exit gate

- Mean and worst-view silhouette, boundary, depth and normal metrics improve without topology regression.
- Unsupported hidden geometry remains flagged rather than scored as correct.
- A face/idol relief is recognisable from geometry without relying on colour texture alone.

Milestone E — component-aware CAD and dependent cavities

Deliverables

- Universal component graph for stones, settings, supports, cavities, metal bodies, links and decorations.
- Individual stone resize/removal/replacement.
- Automatic seat/cavity regeneration from the edited stone geometry and manufacturing clearance.
- CadQuery/OpenCascade exact solids with Blender/Manifold research fallback for difficult Booleans.

Exit gate

- Ten-stone test preserves ten stable IDs.
- Editing one stone changes only its dependent seat/support region.
- Exact-solid validity, collision, wall-thickness and export checks pass.

Milestone F — texture, BRDF and microdetail

Deliverables

- Exposure-normalised, confidence-weighted texture atlas.
- Albedo, material, roughness, normal and displacement outputs.
- NeRO/nvdiffrec experiment on controlled reflective captures.
- Source-camera photometric comparison under fitted lighting and a neutral relighting test.

Exit gate

- Texture seams, baked highlights and view inconsistency are measured.
- Geometry-only and appearance-only ablations are included.
- A different light can be rendered without the original highlight being fixed in the albedo.

Milestone G — manufacturing validation

Deliverables

- Known metric scale and units.
- Minimum wall, clearance, seat, prong and casting rules supplied by the manufacturing partner.
- STEP inspection, mesh manifold checks and scan-to-CAD deviation map.
- Final decision report with accepted, inferred and unknown dimensions.

Exit gate

- No `manufacturing_ready` result without physical scale and independent evidence.
- Scan/CAD deviations are reported in millimetres by component and surface region.

9. Validation matrix

Layer	Metric	Why it matters
Input	Blur, clipping, coverage, overlap, calibration validity	Bad evidence cannot be repaired reliably downstream
Segmentation	Per-instance IoU, Dice, boundary F1, count accuracy	Whole-object IoU can hide missing stones or prongs
Camera	Reprojection error, focal error, pose consistency	Incorrect cameras make all later comparisons misleading
Depth	AbsRel/RMSE against scan where available, cross-view consistency	Measures actual surface distance rather than appearance alone
Normals	Mean/median angular error and edge-aware error	Protects curves, facets and shallow relief

Layer	Metric	Why it matters
Geometry	Chamfer/Hausdorff and signed scan-to-CAD deviation	Finds local shape error hidden by silhouette scores
Visible views	Per-view silhouette IoU and boundary F1	Prevents a strong front view hiding a failed side/back view
Components	Count, stable IDs, pose/dimension errors	Required for editability
Topology	Watertightness, manifoldness, connected bodies, intersections	Required for reliable geometry processing
CAD	Valid B-rep, editable dependency graph, successful STEP round-trip	Distinguishes engineering CAD from a display mesh
Appearance	PSNR/SSIM/LPIPS plus seam and highlight tests	Measures texture without confusing it with geometry
Manufacturing	Scale, wall thickness, clearance, collision, scan deviation	Required before production claims

Every average must be accompanied by the lowest-performing view and lowest-performing component. A missing prong cannot be hidden by a large, accurate shank.

10. Compute plan for the current laptop

The workstation contains an RTX 3050 Mobile-class NVIDIA GPU with 4 GB VRAM and 15 GiB system memory. At this report's system probe, Linux still detected the GPU and loaded NVIDIA kernel modules, but `nvidia-smi` could not communicate with the driver. The root filesystem had 5.6 GiB free and `/home` had 127 GiB free. GPU experiments should not start until driver communication is restored and a fresh VRAM probe succeeds.

Practical local workload

- OpenCV, GrabCut, audit generation and deterministic metrics;
- SAM 2.1 Tiny at controlled resolution;
- the small U-Net and current visual hull;
- Depth Anything V2 Small;
- limited Depth Pro/DSINE inference one image at a time, after memory profiling;
- COLMAP sparse reconstruction and small dense experiments;
- CadQuery/OpenCascade modelling and most topology checks;
- independent preview rendering.

Remote 24 GB or larger GPU workload

- VGGT and larger DA3 variants;
- MAST3R/DUST3R high-resolution multi-view runs;
- NeRO, nvdiffr and Neuralangelo training/optimisation;
- high-resolution differentiable rendering;
- large generative image-to-3D benchmarks.

All remote work should use pinned containers, immutable input hashes and downloaded outputs. Free hosted demos are useful for smoke tests but are not a reproducible training or validation platform.

11. Experiment design and backtracking rules

Each model or algorithm receives an isolated adapter and writes a versioned run:

```
runs/reconstruction/<asset>/<method>/<run_id>/
├─ input_manifest.json
├─ environment.json
├─ cameras.json
├─ depth/
├─ normals/
├─ masks/
├─ geometry/
├─ textures/
├─ renders/
├─ metrics.json
├─ preview_review.json
└─ decision.md
```

The decision is one of `accept`, `reject`, `backtrack` or `research_only`.

A change is accepted only when:

- the intended metric improves on held-out, human-reviewed evidence;
- the lowest view and component do not regress beyond a declared tolerance;
- topology and exact-solid checks remain valid;
- runtime and memory remain practical for the selected deployment tier;
- the licence is compatible with the intended use;
- the actual exported geometry passes independent preview inspection.

Rejected runs remain in the experiment registry with their failure reason. Backtracking is therefore a controlled research action, not loss of progress.

12. Recommended execution order

1. Restore and verify NVIDIA driver communication; keep model caches under `/home`.
2. Create human-reviewed Ring01 and multi-stone masks with stable component IDs.
3. Capture one calibrated, measured jewellery object with both five-view and dense-view subsets.
4. Add the common camera/depth/normal adapter contract.
5. Benchmark COLMAP first as the inspectable geometric baseline.
6. Benchmark Depth Anything V2 Small and DSINE locally.
7. Run DA3, VGGT and MAST3R remotely on identical inputs.
8. Build the uncertainty-weighted fusion and compare five views against dense views.
9. Fit hybrid exact/free-form geometry and run the existing render-and-compare loop.
10. Implement per-stone dependency and cavity updates.
11. Add reflective inverse-rendering and PBR texture baking.
12. Validate against scan/CAD ground truth before setting manufacturing thresholds.

13. Final assessment

Open-source models can provide nearly every technical building block: segmentation, camera estimation, depth, normals, correspondence, reflective inverse rendering, differentiable optimisation, meshing and exact CAD. They do not remove the information limits of the input photographs.

For five ordinary reference images, the defensible output is:

- a high-fidelity visible replica where evidence is strong;
- editable jewellery structure for recognised components;
- plausible but explicitly uncertain hidden geometry;
- non-metric output unless a dimension is supplied;
- no manufacturing claim without independent physical validation.

For five calibrated high-resolution images plus one known dimension, the system can produce a substantially stronger metric and structural result for simple and medium-complexity jewellery. For dense relief, deity faces, complex ornaments, chains, transparent stones and hidden construction, the pipeline should automatically request additional macro/oblique views or scan evidence.

The path to near-perfection is therefore not “find one perfect image-to-3D model.” It is controlled evidence, specialised models, confidence-aware fusion, jewellery constraints, hybrid CAD/free-form geometry, render-and-compare refinement and honest validation.

14. Source register

Primary project and research sources used for this report:

- [COLMAP](#) and [capture guidance](#)
- [OpenMVS](#)
- [VGGT](#) and [CVPR 2025 paper](#)
- [Depth Anything 3](#)
- [Depth Anything V2](#)
- [Depth Pro](#)
- [DUST3R](#) and [MASt3R](#)
- [DSINE](#)
- [SAM 2](#), [HQ-SAM](#) and [Grounded SAM](#)
- [TEED](#)
- [NeRO](#)
- [nvdiffrrec](#), [nvdiffrast](#) and [PyTorch3D](#)
- [Nerfstudio](#) and [Neuralangelo](#)
- [CadQuery](#) and [Blender Boolean documentation](#)

Before a model enters a commercial workflow, review the code licence, checkpoint licence, training-data restrictions and redistribution terms separately. “Open source” or public code does not automatically mean unrestricted commercial use.